



It Takes Two to OЯTHO: A Tabletop Action-Based Embodied Design for the Cartesian System

Katarzyna Potega vel Żabik¹ · Dor Abrahamson^{2,3} ·
Ilona Kłowiecka-Tańska¹

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Abstract

The OЯTHO exhibit, situated within the LivingLab at Copernicus Science Centre (CSC) in Warsaw, Poland, introduces visitors to the Cartesian principles of representing numerical data as points in a two-dimensional space—the coordinate system. This interactive tabletop digital exhibit takes the form of a collaborative two-player game where participants manually control a virtual ball's x - and y -axis values, respectively, to navigate it through maze-like paths, with the objective of completing the course in the shortest time and with the fewest moves. An action-based embodied design, OЯTHO draws on: (1) the enactivist tenet that individuals' cognitive structures emerge from recurring task-effective sensorimotor patterns discovered through explorative perceptuomotor activity; and (2) cognitive-anthropological theorizations of shared ontologies as emerging through multimodal social interaction to facilitate the coordinated enactment of joint action. We overview the exhibit's theoretical underpinnings and design conjectures; detail design challenges particular to the museum context, both in terms of the institution's civic mandate and in terms of attracting and sustaining user engagement; present the exhibit itself; and speculate on the typology of research projects centered on mixed-methods multimodal data analysis based on expected 70,000 annual users.

Keywords Co-operative action · Embodied design · Enactivism · Joint action · Mathematics education · Museum exhibit

Background and Rationale

Interactive technology is carving new frontiers in mathematics education by transforming abstract concepts into concrete and social experiences. This Snapshot introduces the OЯTHO exhibit, a pedagogical interactive technology,

✉ Katarzyna Potega vel Żabik
Katarzyna.potega@kopernik.org.pl

¹ Copernican Revolution Lab, Copernicus Science Centre, Warsaw, Poland

² University of California, Berkeley, USA

³ Korea University, Seoul, South Korea

wherein visitors discover and practice the structural logic of the Cartesian coordinate system through collaborative goal-based play.

Antecedents

Previous design solutions in this genre of interactive museum exhibits include the following: *Geometris*, a geometry learning game that combines elements of *Tetris*TM and *Twister*TM (Rosenbaum et al., 2020); *Math Square*, a multi-user exhibit at the National Museum of Mathematics (MoMath), where visitors collectively navigate mathematical challenges (Shoyfer, 2018); and, closest to home, *Drawing in Motion*, an exhibit developed by the Oregon Museum of Science and Industry, where two participants collaboratively operate a digitized *Etch A Sketch*TM, one participant per rotating knob, to draw images on a screen (Nemirovsky et al., 2013). Inspired by this interactive exhibit genre, OЯTHO offers further empirical context to evaluate the potential educational significance of leveraging technology to foster intuitive embodied understandings of challenging mathematical concepts.

Design Problem

Coordinate systems, such as the Cartesian coordinate plane, are used in many subjects in the fields of mathematics and statistics, natural sciences (e.g., geography, physics), social sciences (e.g., cognitive psychology), and humanities (e.g., history). Correctly using Cartesian coordinate systems is essential to understanding graphs, maps, and models. However, many students have difficulties analyzing graphs (Bell & Janvier, 1981; Kerslake, 1981; Leinhardt et al., 1990) and reading maps (Downs & Liben, 1991). In particular, Sarama et al. (2003) diagnose 4th grade students' challenges with coordinate systems as follows:

order and distance relationships....must be constructed and coordinated across the two dimensions. Labels must be related to grid lines and, in the form of ordered pairs of coordinates, to points on the grid, and eventually integrated with the grid's order and distance relationships so that they constitute a mental number and ultimately can be operated upon. (p. 288)

We were therefore motivated to ideate the design of an activity that would occasion opportunities for students to: (a) become oriented in the Cartesian coordinate system, thus perceiving a point in two-dimensional space as cross-plotted at the intersection of two intersecting orthogonal lines that emanate, respectively, from horizontal and vertical reference axes; and (b) understand and correctly use the conventional rules for expressing and symbolizing a point on the plane as an ordered pair, for example “(5, 3),” and, vice versa, to locate a point marked as such.

Design Challenges Particular to the Science Centre Context

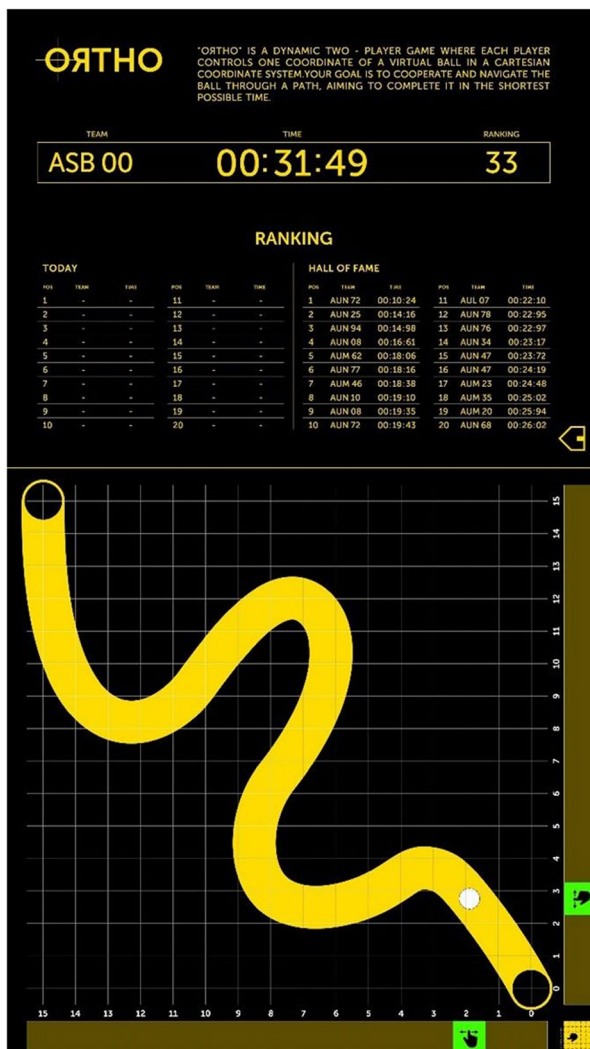
The intended placement of the OЯTHO exhibit within the context of the science center exhibition, unlike school or other educator-facilitated activities, presented distinct challenges that influenced the initial design criteria:

1. **Self-contained information:** The exhibit should be purposefully crafted to provide all essential information within the interaction. This design choice aimed to cater to the diverse array of visitors who would engage with the exhibit independently, without the need for external guidance or explanations.
2. **Engagement:** Given the competition for visitors' attention among 450 other exhibits, OЯTHO had to be engaging and captivating to stand out. One way of drawing visitors was to make the interaction necessarily collaborative. Another quality of activities known to attract visitors is to deploy principles of gamification, including a ranking system that encourages competition and motivation among participants.
3. **Minimal text:** Acknowledging the presence of diverse visitors, including children from multiple nationalities (e.g., Ukrainian refugees) with varying reading abilities, we aimed to minimize user reliance on textual instructions. Instead, we resorted to visual cues, such as elements that illuminated or changed color, that were to serve as intuitive guides throughout the interaction (see Fig. 1). Additionally, when deliberating over how best to convey the game's rules, a conscious decision was made to abstain from employing textual instructions. Instead, each "game" played would conclude upon making a mistake (i.e., the digital ball falls off its path—"game over"). This approach was intended not only to streamline the learning process but also to facilitate understanding.

Design Solution

Working in the embodied design paradigm (Abrahamson, 2014; Abrahamson et al., 2024), we strived to model the activity on two ideation heuristics: (1) the enactivist tenet that individuals' cognitive structures emerge from recurring task-effective sensorimotor patterns discovered through explorative perceptuomotor activity (Varela et al., 1991); and (2) cognitive-anthropological theorizations of shared ontologies as emerging through multimodal social interaction to facilitate the coordinated enactment of joint action (Goodwin, 2013). These heuristics might be paraphrased more colloquially as stating that: (1) when you try to do something new, like tying your shoelaces for the first time, as you grope for good holds that let you get the job done, you might come to see and feel the world in a new way, such as noticing embedded structures and figuring out how to grip or create them; and (2) if you are doing the job with another person, you will want to figure out ways of referring to things by pointing to them, naming them, and telling the other person what to do with those things, so that a new local situated language will pop up that serves you two as a resource both now and moving forward. For theoretical coherence, we hasten to note that the enactivist notion of cognitive structures emerging from recurrent

Fig. 1 Screenshot of the “OЯTHO” game, Level 1. The two players are positioned along orthogonal axes. Each player controls the motion of the white ball only along their respective axis (see green “hand” icons), such that diagonal ball trajectories along the narrow yellow pathway require careful coordination of simultaneous distributed action, which typically engenders the emergence of proto-mathematical language



task-effective sensorimotor exploration and the cognitive-anthropological notion of discerning feature structures in the environment are two sides of the same non-dualist epistemic coin. Neither the cognitive structures are “in the head” nor are the feature structures “in the environment.” Rather, adaptive neural changes occur through/for/ with the perception of new Gestalts in the environment to enable these contextually operable perceptions henceforth. As Piaget (1971) famously stated, “Knowing does not really imply making a copy of reality but, rather, reacting to it and transforming it (either apparently or effectively) in such a way as to include it functionally in the transformation systems with which these acts are linked” (p. 6). Finally, collaborative joint action is conducive to reflecting on one’s own skill and articulating it in multi-modal language utterances. You thus come to know what you know to do.

Our design, which was initially sketched by the second author and then co-developed by the CSC team, resulted in an interactive two-player game, “OЯTHO,” where participants collaboratively control a virtual ball’s coordinates to navigate it through maze-like paths, with the objective of completing the course in the shortest time and with the fewest moves. Earlier generations of this design’s general mechanics—distributed orthogonal control of a digital object’s movement across a Cartesian field—can be found in the Mathematics Imagery Trainer research at the Embodied Design Research Laboratory, University of California, Berkeley (Lee et al., 2013), and with their collaborators at Utrecht University (Shayan et al., 2015).

Broader Context—LivingLab

The “OЯTHO” exhibit is part of a group of exhibits situated within the dedicated area of the Copernicus Science Centre’s exhibition known as the LivingLab. This concept was conceived to facilitate a three-way dialogue between the science centre, visitors, and scientists, forging a collaborative environment where science and discovery converge. The Copernicus Science Centre (CSC) provides a physical platform for this unique endeavor, allocating exhibition space for five distinct research stations, including “OЯTHO.” This commitment represents a transformative stride toward redefining the landscape of scientific communication. By opening its doors to researchers and their projects, CSC embarks on a journey of research collaboration, fostering a new dimension of engagement between science and society. Within the LivingLab framework, visitors can actively engage with science by undertaking tasks and challenges presented at the various research stations. This interactive participation structure is intended to allow visitors to grasp the intricacies of scientific knowledge production and experience a sense of accomplishment and self-discovery. For scientists, the LivingLab should serve as an invaluable arena for the design and execution of tasks and experiments. Moreover, scientists are invited to contribute toward cultivating social engagement in research projects, thus fostering a sense of shared responsibility and collaboration in scientific exploration. The LivingLab concept exemplifies a pioneering approach to science education and engagement. Visitors can actively participate in the scientific process through exhibit-like interactive research stations designed to provide them with insights into the dynamics of knowledge production. Since its installation in July 2023, the “OЯTHO” exhibit has consistently attracted approximately 100 pairs of visitors daily.

The LivingLab initiative draws inspiration from projects like “Science Live!” at the NEMO Science Museum in Amsterdam, where researchers, mainly from the social and behavioral sciences, are invited to conduct experiments in a laboratory embedded within the exhibition area. Similarly, “NanoToTouch,” an EU-funded project from 2009 to 2011, saw several museums across Europe establish open laboratories on their exhibition floors. Here, visitors could observe researchers at work, primarily from the fields of science and technology, and engage in conversations with them.

The OЯTHO Exhibit

The OЯTHO game integrates an interactive module equipped with a Naos+ 86" Multi-Touch Display and operates on Microsoft Windows 11 Pro PL 64-bit OEM, marked by product ID FQC-10544. Intended for educational exhibition at museum learning stations, the unit utilizes software programmed by Unique Interactive designed to gather data on user interaction with the exhibit described below.

Description of the Interaction At the outset of the interaction, players are presented with information regarding the game and the ongoing research. Participants are asked to consent to participate in the research study. Following this, we request participants to share some basic information about themselves—their age and the nature of their relationship (whether they are familiar with each other). The gameplay then begins. Upon completing a level, players can either attempt the same level again for further practice/better results or advance to the next level. OЯTHO offers multiple maze layouts for each level, which are randomly generated. Each level offers a distinct experience, gradually increasing complexity and requiring players to adapt and collaborate effectively. Here, we provide an overview of the gameplay progression through each level.

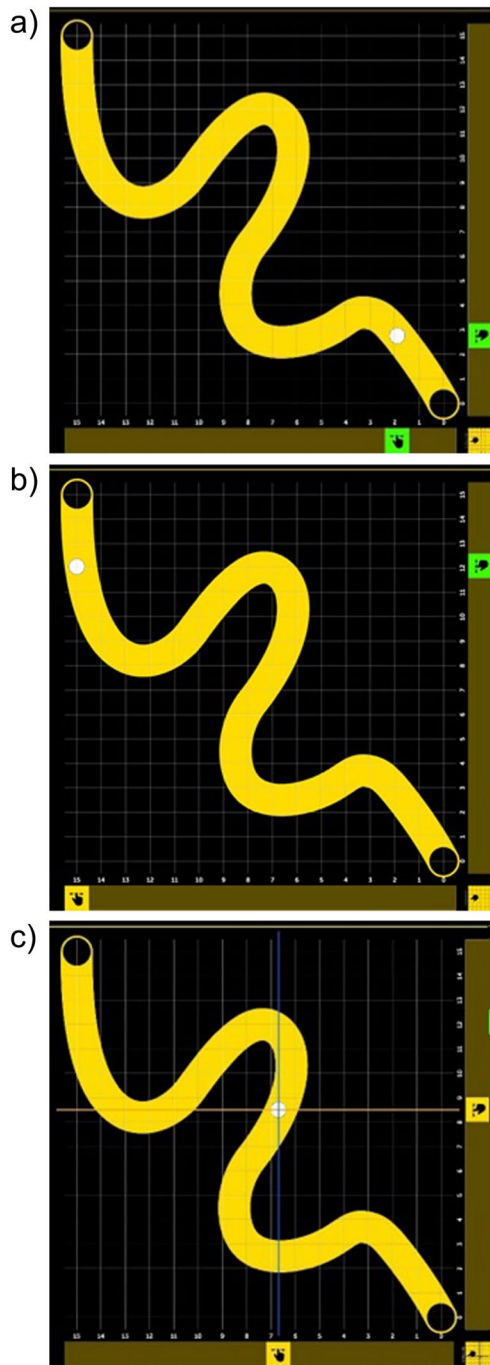
Level 1: Cooperative Coordination

The game commences with a countdown, giving participants a brief moment to acquaint themselves with its mechanics. At this level, two players maneuver the digital ball from the starting point (coordinates 0, 0) to the designated “hole” (see Fig. 2—a circle in the top-left corner). However, player control is limited to horizontal (left–right) or vertical (up–down) movement. Players navigate the virtual ball by touching and moving the hand icons along the navigation strips placed at the board’s edges. If the ball moves off the path, it’s “game over.” As such, from the get-go we deploy both of our design heuristics into the collaborative game mechanics: (1) players engage in explorative sensorimotor activity so as to enable their perception to guide their action; even as they (2) coordinate their moves with their game partner so as to effectively enact the joint action necessary to maneuver the ball along its designated path.

Players can enable the crosshairs to assist in precision movements as needed (see Fig. 2—icon in bottom right corner). Even though the grid and the coordinates are not required at this level to complete the task, we have included these features for two reasons. Firstly, the grid is a valuable reference point during player negotiations, aiding in effective communication and coordination. Secondly, its presence is essential in maintaining continuity, as it becomes indispensable in the subsequent two exhibit levels. After the successful completion of the task, users are presented with their score and, if they so wish, can repeat this level and try to improve their time, or they move on to the second level.

Fig. 2 Level 1 board features.

a Beginning of interaction without crosshairs feature on.
b Ending the interaction without crosshairs.
c Interaction with crosshairs feature on



Level 2: Introduction of Coordinates—Transitioning from Manual to Numerical Control

Level 2 is a pivotal phase in bridging the semiotic gap between manual control and the utilization of numerical representations within the OЯTHO exhibit. In this stage, players experience a shift in gameplay dynamics, marking the evolution of their control method. Now, they must each set a crosshair through their intended destination and then one of them activates movement by pressing the “GO!” button (see Fig. 3, bottom-right corner). Upon pressing the button, the ball moves in a straight line to the point indicated by the crosshairs; if the ball exits the path area, the game is over. This level introduces a crucial juncture, where players adapt to a more precise and strategic approach, adding an element of numerical control to the game. To aid in clarity, each player’s corresponding coordinate number is visible at the intersection point, color-coded to match the crosshair they control (see Fig. 3), thus indicating the practical semiotic relation between the two display elements. Whereas the coordinate numbers (the ordered pair) are not strictly necessary for navigation, the authors made the conscious decision to add the numbers as an introduction to the next stage of the game. Finally, the scoring system considers an additional metric—the number of movements executed.

Level 3: Full Numerical Control

The gameplay takes a significant turn in the third level, as the navigation strips transform. Numeric values replace the visual aids (hand icons and guiding crosshairs), necessitating a shift from manual manipulation to numerical input. In this challenge, players must use the numerical values displayed on the navigation strip to determine and set the coordinates for the ball’s sequence of interim destinations along its path. Once both players agree on the coordinates and have pressed their respective numeric buttons (see Fig. 4—round yellow buttons, with numbers 0 to 15, placed along the edges of the game field), they press the “GO!” button to set the ball in motion. Similar to the game mechanics in Level 2, the ball moves in a straight line to the location indicated by the coordinates set by the users. This level, therefore, demands effective verbal communication and cooperation from the players for successful navigation—they must agree on the coordinates of the destination where the ball will be sent. The level introduces a distinctive feature, where coordinates (the ordered pair) appear at each destination to which the virtual ball was directed (see Fig. 4). Notably, these coordinates remain visible on the board throughout the interaction. With this, we have completed explaining the three-leveled activity.

Description of the Data Structure Data are being collected from each interaction and compiled into CSV-format files from all interactions throughout the day. The exhibit interaction data are composed of several elements that include participation preliminaries—date and time, consent status, language preference selection, and team identification code—as well as logged timestamps, with millisecond precision, of all digital interactions (i.e., all icon manipulation, button presses, numerical input, etc.),

Fig. 3 Level 2 board. **a** Beginning of interaction. **b** Crosshairs placed in the desired position, and its ordered pair appears. **c** The ball has moved to the selected position

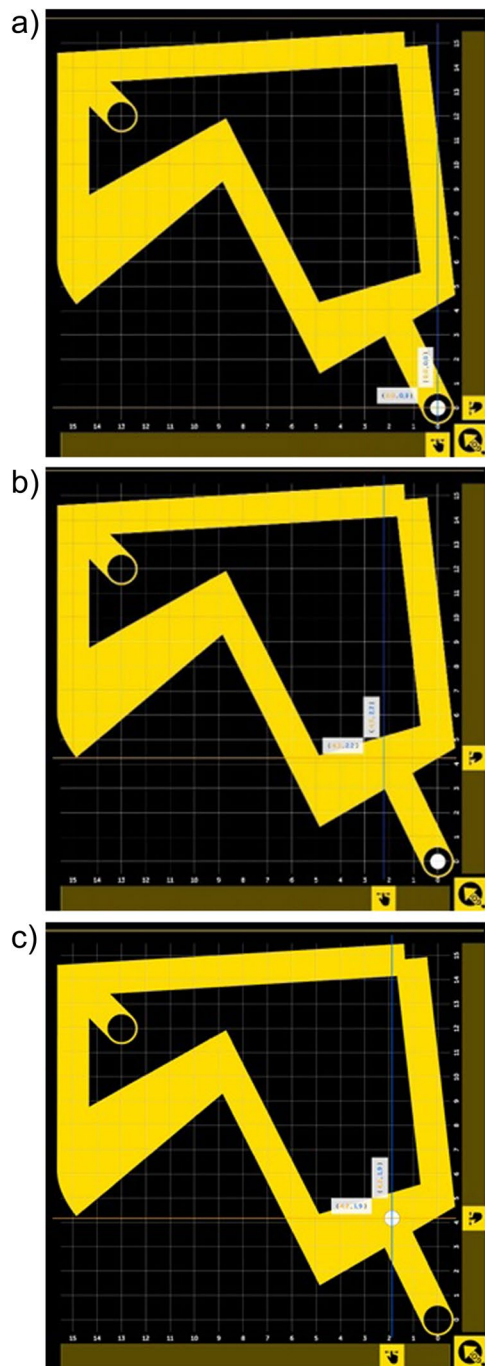
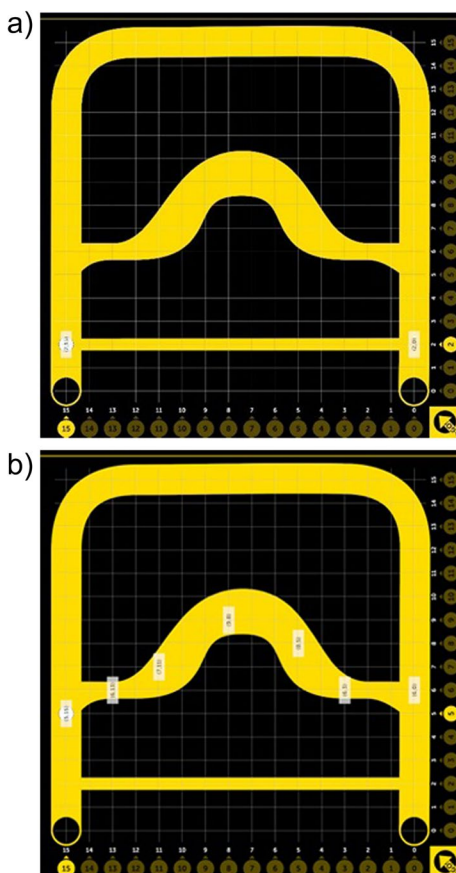


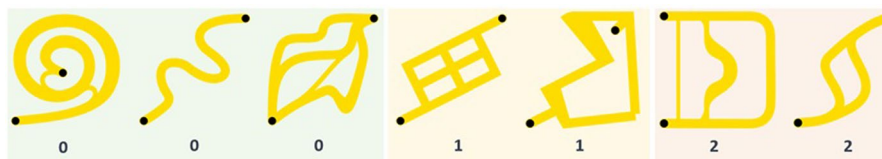
Fig. 4 Level 3 board. **a, b**

Examples of two different routes taken, with different numbers of moves made between Start and End. Note that the manual control screen element is no longer a continuous slider but rather a sequence of numerical values



from which performance parameters can be directly derived. The dataset further indicates whether or not the team completed all three levels. Seven track identifiers and associated three levels are noted (see Fig. 5), with timestamps marking the start and end of each level attempt, along with the duration in milliseconds.

Data on track completion, errors (e.g., the ball driven off the track), and level completion are registered. The times at which specific buttons are pressed and the visibility status of the crosshairs are recorded. The position of the moving ball is

**Fig. 5** Seven sample path items grouped into three levels (0, 1, 2)

recorded as X and Y coordinates every 16.67 ms (frequency 60 Hz), which provides insight into the interaction dynamics. The position is measured in pixels, where the origin point (0, 0) in the coordinate system corresponds to the value 0, 0 in pixels, and the point (15, 15) in the coordinate system corresponds to the value 1800, 1800 in pixels. The dataset reflects the ball's status, indicating whether it is in its initial position, has been relocated correctly, has reached the track's end, is off the track due to an error, or has been manually repositioned to a new coordinate following a mistake. This capture of structured data allows for detailed subsequent analysis.

Research: Preliminary Observations and Future Plans

ORTHO offers an auspicious platform for empirical research into multiple socio-cognitive phenomena germane to the learning sciences agenda of promoting educational theory and practice. In particular, ongoing onsite research at CSC with local and international collaborators should yield future insight into the microgenetic processes by which mathematical discourse emerges and is linguistically negotiated. Previous studies with an activity generally similar to ORTHO have characterized students' learning trajectory from sensorimotor exploration to mathematical discourse by noting how they adopted available frames of reference, such as a grid of lines, to enhance the enactment, evaluation, or explanation of their solution strategy (Abrahamson et al., 2011). However, that line of research included scant cases of dyadic work. We are particularly interested in investigating the process by which two people figure out how to engage in task-oriented collaborative action in unfamiliar digital territory. For example, we will examine whether and how players establish game-based ethnomethods, that is, mutually intelligible routines for mobilizing smooth joint action. We conjecture that players will generate, coordinate, and integrate two aspects of their manual actions: (1) *functional*—steering the ball along its path toward the destination, each player contributing their axial vector; and (2) *semiotic*—communicating intentions by modulating nuances of action parameters, such as onset, speed, acceleration, direction, and distance from path center. Modeling these behaviors more generally should prove useful for our broader research agenda of building and evaluating engaging collaborative STEM learning activities, while contributing to the learning sciences literature on socio-cognitive theory.

Data collection is ongoing, storing records of users' demographic input, digital actions, and—occasionally, from the activity of selected volunteers—video–audio data, eye-gaze tracking data, and field notes. Meanwhile, field observations on site, follow-up conversations with visitors, and preliminary analyses all indicate that visitors swiftly comprehend the exhibit's core objectives and game mechanics by investigating and deducing cause-and-effect relationships between their actions and the game's outcomes. Furthermore, ORTHO's visitor-centric design appears to have seamlessly integrated the exhibit into the environment of the CSC, providing an accessible, engaging, and self-explanatory learning experience for all visitors.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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